Report Maersk Developer Oil Rig

Bürli Norman, Frongillo Matteo, Spichtig David



HS26 – Corporate Ethics and Sustainability – Prof. Dr. Claas Wagner

Question

1. Describe your initial operating strategy, covering the following points:
    - Production volume
    - Overhauls
    - Maintenance and safety spending
    - Crew management

Initial strategy

Production volume

In practice round 1 (PR1), the production volume was tested by increasing the production to its maximum (1,650,000 bbls) and hiring more workers. This allowed the group to understand how extreme conditions affect the sustainability factors.
The takeaway for PR1 was that increasing the production volume to its maximum value was not the correct strategy under the specific starting conditions of PR1, as profit was reduced by high hiring costs.

In practice round 2 (PR2), crude oil production was not prioritized; instead, safety was prioritized.

Decisions in PR1 and PR2 were made to understand how the economic factors respond to these decisions.

The production volume in Round 1 was set to 1,300,000 bbls. To ensure a high volume, ten workers were hired. The same strategy was applied for Round 2.

After Round 1 and Round 2, the group was breaking even. This result is helpful in identifying the threshold at which production becomes profitable under specific conditions.

The next steps in production aim to keep the company profitable.

Overhauls

The overhaul schedule in Figure 1 illustrates how the timing of major overhauls affects the BOP value over successive rounds.

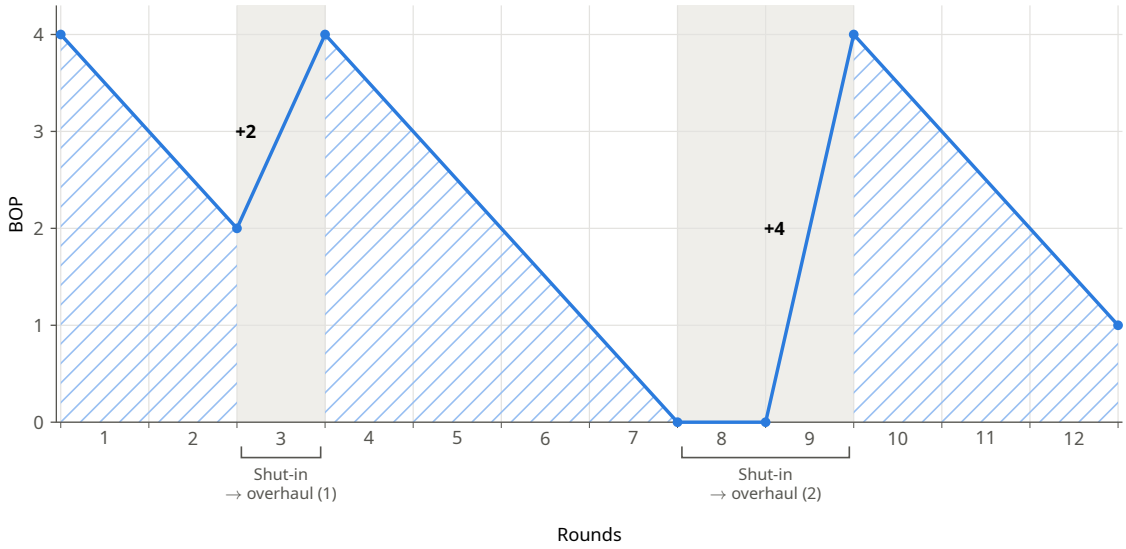


Figure 1: BOP planning schedule

The weather report predicts Beaufort 10 (Storm) in Round 3. For safety reasons, a shut-in is necessary if a storm occurs, as the platform and the oil pumps might be compromised. To mitigate the risks, an early overhaul is the best option because it restores the BOP before the expected storm, reducing the risk of a blowout. Assuming that a Beaufort 10 will occur only in Round 3, this strategy allows crude oil production to continue for multiple rounds, meaning that major overhauls would only be needed in Rounds 8 and 9.

The group's strategy is that the oil rig should never go below 0 BOP, as the company's priority is safety, and falling below the BOP threshold would increase the risk of a blowout, carrying out major overhauls at six-round intervals, regardless of the BOP value.

Maintenance and safety spending

The company's main goal is to ensure the safety of the oil rig by consistently investing in maintenance and keeping the Rig Equipment Condition Index (RECI) above its baseline value of 100.

To achieve this goal, two main strategies are being considered:

1. Invest on maintenance and safety to keep the probability of failures and accidents low
   → Expensive but safe

Table 1: Maintenance and safety expenditure

	Maintenance [$/bbl.]	Safety [$]	RECI
Baseline	8	1,200,000	100
Round 1	12	2,000,000	141
Round 2	12	2,000,000	182

2. Let the RECI degrade and restore it to its baseline value (100) during major overhauls perfomed every six rounds.
   → Cheap but risky.

During Rounds 1 and 2, the first strategy was used. To decide which strategy to adopt, a better understanding of the RECI and its effect on safety is needed.

Crew management

Because one of the goals is to ensure the rig's safety, five crew members are hired each round. This strategy allows the company to increase production volume while keeping the tiredness index low.

Each round, five crew members receive training, allowing expertise to develop over time while keeping training costs low.

Ethical Questions

#1: Arctic Investment Opportunity

The Arctic investment was considered too risky given the expected return and the high initial cost. Therefore, no investment was made.

#2: Backup Battery

The BOP backup battery test costs approximately $5 million.

Table 2: Cost and probability comparison of a blowout and the BOP backup battery test

	Cost [1e6 $]	Estimated chance [%]	Normalized cost [1e6 $]
Blowout (minimum)	250	1	2.5
Backup Battery test of BOP	5	100	5
Blowout (realistic)	500	1	5

A blowout has an estimated minimum cost of $250 million, while a more realistic estimate is $500 million. Assuming a 1% probability of blowout, the expected cost would be $5 million, equal to the cost of the test (Table 2).

Ensuring safety is a priority of the company’s strategy, so based on these assumptions, carrying out the test is also justified, as it helps verify the reliability of the backup system and supports safe operation and continued production.